λ

Type systems, errors, variables and functions

Programmazione Funzionale

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Università di Trento

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Tutoring

- Starting from next Tuesday
  - 17:30 – 18:30 in PC A202
  - Except 17-03-2026 17:30 – 18:30 in PC A201

- If you need, you can reserve a slot on Friday morning (10:30 – 11:30) in Moodle

Today

- Type systems
- Type errors and conversion in ML
- Names, denotable objects and variables
- Variables in ML
- Complex types in ML
- Functions in ML

Type systems

Type systems

- The type system of a language:
  1. Predefined types
  2. Mechanisms to define new types
  3. Specification of whether types are statically or dynamically checked
  4. Control mechanisms
     - Equivalence
     - Compatibility
     - Inference

Simple and composite types

- Simple (or scalar) types: types whose values are not composed of aggregations of other values
- Composite types: obtained by combining other types using appropriate constructors, e.g., records, vectors, sets, pointers
- We define new types for example with

```
type newtype = expression;
```

Type systems

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Static and dynamic type checking

Static type checking

- At compilation time (C, Java, Haskell, ML)
- Pros
  - At compilation time, before going to the user
  - It is efficient at runtime
- Cons
  - Design is more complex
  - Compilation takes longer
  - More conservative: static type errors are not runtime errors

```
int x;
if (0==1) x="pippo";
```

Dynamic type checking

- During code execution (Python, Javascript, Scheme)
- Pros
  - it locates type errors
- Cons
  - It is not efficient
  - The type error is identified only at runtime with the user

Strongly and weakly typed languages

Strongly typed

- informally, languages that are strict about types
- the type of a value does not change in an unexpected way (e.g., implicit type conversions are not allowed)

ML

```
> 4+"2";
poly: : error: Type error in function application.
   Function: + : int * int -> int
   Argument: (4, "2") : int * string
   Reason:
      Can't unify int (*In Basis*) with string (*In Basis*)
         (Different type constructors)
Found near 4 + "2"
Static Errors
```

Weakly typed

- informally, languages that are more relaxed about types
- the type of a value can change in an unexpected way (e.g., implicit conversions are allowed)

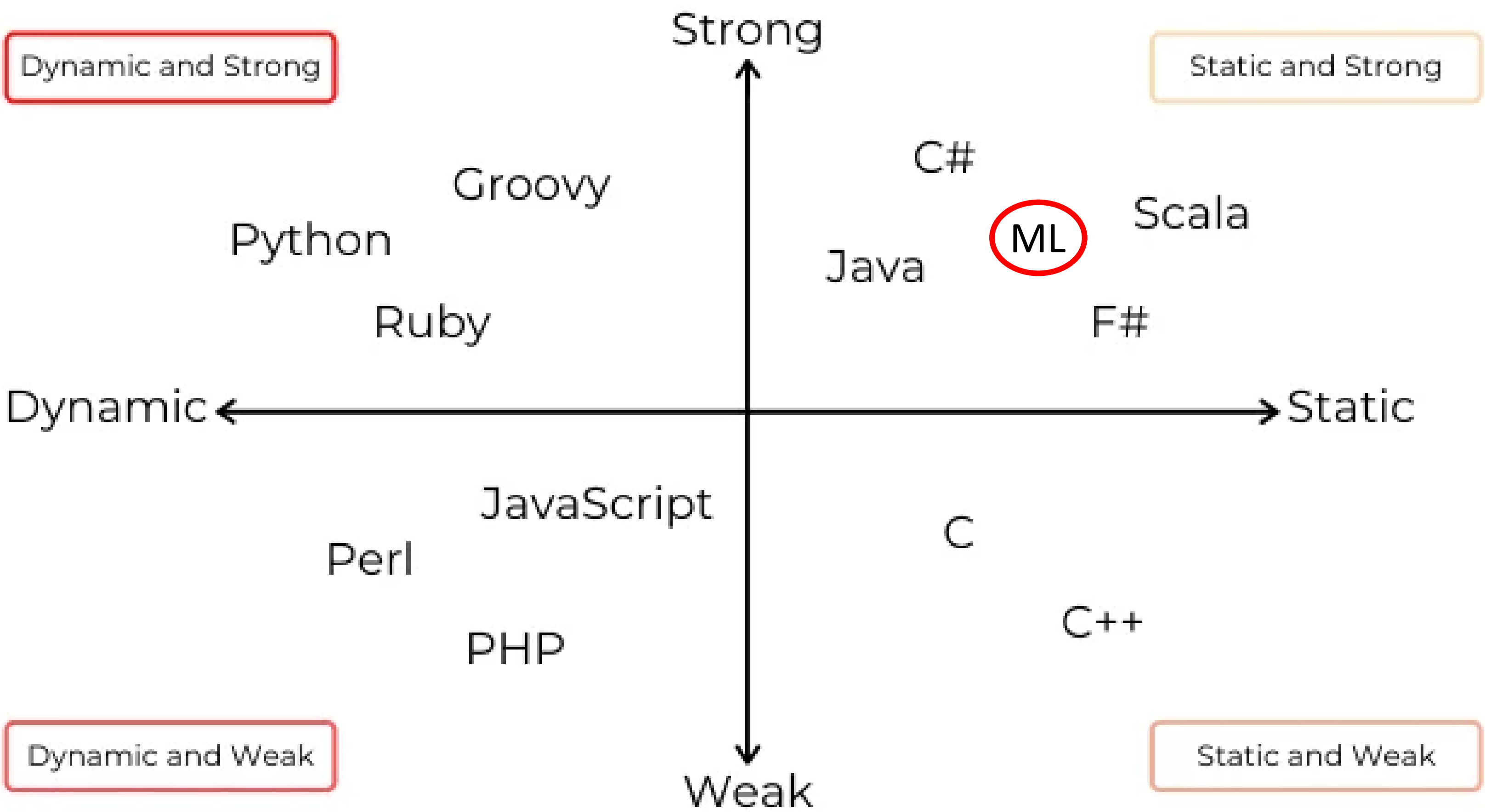
Javascript

```
4 + '2'
> '42'
```

Php

```
<?php $str = "candies";
$str = $str + 10;
echo ($str); ?>
> 10
```

Strong, weak, dynamic and static



Strong typing is an aspect of type safety

[Prepinsta.com]

Type safety

- A type system is type safe if
  - no program can have undetected errors deriving from type errors
- Type safety features ensure that the code does not perform any invalid operation on the underlying object
  - type error checking can be carried out at compile time or at runtime
- A strongly typed language has a high degree of type safety

C, C++ are type unsafe

- For instance, C and C++ do their best for accommodating casting from one type to another.

```
void func(char* char_ptr) {
    double* d_ptr = (double*) char_ptr;
    (*d_ptr) = 5.0;
    cout << "Value of pointer after cast in func(): " << *d_prt <<
endl; }
```

- The program will claim memory for a double rather than for a char.
- Similarly when allowing the programmer having control over memory allocations

```
int buf[4];
buf[5] = 3; /* overwrites memory */
```

Type systems

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Type equivalence

Type equivalence

- Two types `T` and `S` are equivalent if every object of type `T` is also of type `S`, and vice versa
- Two rules for type equivalence
  - Equivalence by name: the definition of a type is opaque
  - Structural equivalence: the definition is transparent

Equivalence by name

- Two types are (strongly) equivalent by name if they have the same name (Java)

  Which of the following are strongly equivalent by name?

```
type T1 = 1..10;
type T2 = 1..10;
type T3 = int;
type T4 = int;
```

- None of the four
- Too restrictive

- Loose or weak equivalence by name (Pascal)
  - A declaration of an alias of a type generates a new name, not a new type
  - T3 and T4 are names of the same type
- Defined with reference to a specific program, not in general

Structural Equivalence

- Two types are structurally equivalent if they have the same structure: substituting names for the relevant definitions, identical types are obtained.
- Structural equivalence between types is the minimal equivalence relation that satisfies:
  - A type name is equivalent to itself
  - If `T` is defined as `type T = expression`, then `T` is equivalent to `expression`
  - Two types constructed using the same type constructor applied to equivalent types, are equivalent

Structural Equivalence Examples

Which of the following are structurally equivalent?

```
type T1 = int;
type T2 = char;
type T3 = struct{
    T1 a;
    T2 b;
}
type T4 = struct{
    int a;
    char b;
}
```

```
type S = struct{
    int a;
    int b;
}
type T = struct{
    int n;
    int m;
}
type U = struct{
    int m;
    int n;
}
```

- Some aspects are clear
  - T3 and T4 are structurally equivalent
- Other aspects are less clear
  - S, T and U have field names or order that are different: are they equivalent?
  - Usually no, yes T and U for ML.
- Defined in general – not specifically for a program

Type equivalence in languages

- Combination or variant of the two equivalence rules
  - Pascal → weak equivalence by name
  - Java → equivalence by name – except for arrays with structural equivalence
  - C → structural equivalence for arrays and types defined with typedef but equivalence by name for records and unions
  - ML → structural equivalence except for types defined with `datatype`

Compatibility and conversion

Compatibility

- `T` is compatible with `S` if objects of type `T` can be used in contexts where objects of type `S` are expected
- Example: `int n; float r; r=r+n` in some languages
- In many languages compatibility is used for checking the correctness of:
  - Assignments (right-hand type compatible with left-hand),
  - parameter passing (actual parameter type compatible with formal one), …
- Compatibility is reflexive and transitive but it is not symmetric
  - E.g., compatibility between `int` and `float` but not viceversa in some languages

Compatibility

- The definition depends on the language.
- `T` can be compatible with `S` if
  - `T` and `S` are equivalent
  - The values of `T` are a subset of the values of `S` (interval)
  - All the operations on values of `S` can be performed on values of `T` (extension of record) – sort of subtype
  - There is a natural correspondence between values of `T` and values of `S` (`int` to `float`)
  - The values of `T` can be made to correspond to some values of `S` (`float` to `int` with truncating, rounding)

Type conversion

- If `T` is compatible with `S`, there is some type conversion mechanism.
- The main ones are:
  - Implicit conversion, also called coercion. The language implementation does the conversion, with no mention at the language level
  - Explicit conversion, or cast, when the conversion is mentioned in the program

Coercion

- Coercion indicates, in a case of compatibility, how the conversion should be done
- Three possibilities for realizing the type conversion. The types are different, but
  - Same values and same storage representation for values of T$\subseteq$ S. E.g., types that are structurally the same, but have different names
    - Conversion only at compile time → no code to be generated
  - Different values, but the common values have the same representation. E.g., integer interval and integer
    - Code for dynamic control when there is an intersection
  - Different representations for the values. E.g., reals and integers
    - Code for the conversion

Cast

- In certain cases, the programmer must insert explicit type conversion (C, Java: cast)

  `S s = (S) T`

  For example `r = (float) n` and `n=(int) r`

- Cases similar to coercion
- Not every explicit conversion is allowed
  - Only when the language knows how to do the conversion
  - Can always be done when types are compatible (useful for documentation)
- Modern languages prefer cast to coercion.

Type errors and conversion in ML

Type errors in ML

Type errors

```
> 1 + 2;
val it = 3: int

> 1.0 + 2.0;
val it = 3.0: real

> 1 + 2.0;
poly: : error: Type error in function application.
Function: + : int * int -> int
Argument: (1, 2.0) : int * real
Reason:
Can't unify int (*In Basis*) with real (*In Basis*)
(Different type constructors)
Found near 1 + 2.0
Static Errors
```

What does the error message mean?

Type errors

- Type of + is `int * int -> int`
- Actually, + can have different types, but, based on the first argument, ML decides on the integer version
- What if we have a real as "first argument"?

```
> 1.0 + 2;
poly: : error: Type error in function application.
Function: + : real * real -> real
Argument: (1.0, 2) : real * int
Reason:
Can't unify int (*In Basis*) with real (*In Basis*)
(Different type constructors)
Found near 1.0 + 2
Static Errors
```

- In this case, ML decides that we want real addition: `real * real -> real`
- In both cases, the second argument does not match the first

Other type errors

```
> #"a" ^ "bc";
poly: : error: Type error in function application.
Function: ^ : string * string -> string
Argument: (#"a", "bc") : char * string

> 1/2;
poly: : error: Type error in function application.
Function: / : real * real -> real
Argument: (1, 2) : int * int

> if 1<2 then #"a" else "bc";
poly: : error: Type mismatch between then-part and else-part.
Then: #"a" : char
Else: "bc" : string
```

Type conversion in ML

How to deal with these issues?

Type conversion

Conversion between integers and reals

- From integers to reals
  - `real`: convert from integer to real

- From reals to integers → four possibilities:
  - `floor`: round down
  - `ceil`: round up
  - `round`: nearest integer
  - `trunc`: truncate

Values halfway between two integers go towards the closest even value.
For example:
> round(3.5);
val it = 4: int
> round(4.5);
val it = 4: int

Examples

```
> real(4);
val it = 4.0: real
> real 4;
val it = 4.0: real

> floor(3.5);
val it = 3: int
> ceil(3.5);
val it = 4: int
> round(3.5);
val it = 4: int
> trunc(3.5);
val it = 3: int

> floor(~3.5);
val it = ~4: int
> trunc(~3.5);
val it = ~3: int
```

`trunc` and `ceil` are different on positive values, while `trunc` and `floor` on negative values.

Conversion between characters and integers

- `ord`: from character to integer
- `chr`: from integer to character

```
> ord #"a";
val it = 97: int
> ord #"a" - ord #"A";
val it = 32: int

> chr 97;
val it = #"a": char
```

Integers in the range 0 to 255. Outside the interval Exception – Chr raised

Conversion from characters to strings

- `str`: from character to string

```
> str #"a";
val it = "a": string
```

In other words …

- No automatic conversion of types
  - 5+7 and 5.0+7.0 are correct, resulting in `int` and `real`
  - 5+7.0 is wrong
- Conversion between types
  - `real`: integer to real
  - `ceil, floor, round` and `trunc`: real to integer
  - `ord`: character to integer
  - `chr`: reverse direction, i.e., integer to character
  - `str`: character to string

Names, Denotable Objects and Variables

What is a name?

- Sequence of characters used to *denote* something else
  - `const p = 3.14` (object denoted: the constant 3.14)
  - `x` (object denoted: a variable)
  - `void f()(...)` (object denoted: the definition of `f`)
- In programming languages, the names are often identifiers (alphanumerical tokens)
- The use of a name serves to indicate the object to denote
  - Symbolic identifiers, that are easier to remember
  - Abstractions (data or control)

Names and denotable objects

- A name and the object it denotes are not the same thing.
  - A name is just a character string
  - Denotation can be a complex object (variable, function, type, etc.)
  - A single object can have more than one name ("aliasing")
  - A single name can denote different objects at different times
- We use "the variable `fie`" or "the function `foo`" as abbreviations for "the variable with the name `fie`" and "the function with the name `foo`"

What is a denotable object?

- An object that can be associated with a name
- Names defined by the user: variables, formal parameters, procedures, types defined by the user, labels, modules, constants defined by the user, exceptions
- Names defined by the language: primitive types, primitive operations, predefined constants
- Binding: association between name and denotable object

Environment

- Not all the associations between names and denotable objects are fixed once and for all.
- Environment: the collection of associations between names and denotable objects that exist at runtime at a specific point in a program and at a specific moment in the execution
- Declaration: a mechanism (implicit or explicit) that creates an association in an environment.

```
int x;
int f(){
    return 0;
}
type T = int;
```

Variable declaration

Function declaration

Type declaration

Names and denoted objects

- The same name can denote different objects at different positions in the program

```
{int fie;
 fie = 2;
     {char fie;
          fie = a;
     }
}
```

Integer

Char

- A single object is denoted by more than one name in different environments
  - A variable passed by reference to a procedure

```
procedure P (var &X:integer);
begin
...
end;

var A:integer;
P(A);
```

X

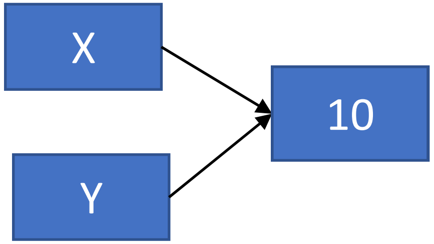
A

Names and denoted objects

- Aliasing: different names that denote the same object in the same environment
  - Pointers (integer pointers)

```
int *X, *Y; // X,Y pointers to integers
X = (int *) malloc (sizeof (int)); // allocate heap
memory
*X = 5; // * dereference
Y = X; // Y points to the same object as X
*Y = 10;
write(*X);
```

10



Variables

- Variable: in mathematics, an unknown that can take values from a predefined domain (that cannot be modified anymore)
- In computer science it depends on the paradigm
- In classical imperative languages (Pascal, C, Ada, etc.,):
  - modifiable variable: the value can be modified
  - A container of values that have a name

`x`	3	The variable with name `x` The variable `x`

Different models

- In some imperative languages (object-oriented ones)
  - a variable is a reference to a value, which has a name and is stored in the heap
  - similar to pointers but without the possibility to directly accessing the location
- In logical languages
  - a variable can be modified only under certain conditions (instantiation)
- In pure functional languages (Lisp, ML, Haskell, SmallTalk)
  - a variable is an identifier that stands for a value (not modifiable), as in mathematics

Variables in ML

Variables

- Environment: Set of pairs of identifiers and values
- It is possible to add an identifier to the environment and bind it to a value
- Environment is modified by val-declarations (a sort of assignment)

```
val <name> = <value>;
val <name>:<type> = <value>;
val <name> = <expression>;
```

- Example

```
> val pi = 3.14159;
val pi = 3.14159: real
> val v = 10.0/2.0;
val v = 5.0: real
```

- Note that the response has the name of the variable, rather than it.

Variables

- We do not need to specify the type
- Variables cannot be modified!
- `val` creates an association between a name and a variable
- A new declaration creates a new variable, and does not change the value of the existing one

```
> val pi = 3;
val pi = 3: int
> val pi = 3.14159;
val pi = 3.14159: real
```

pi	3.14159
pi	3
…	…

create two variables with name `pi`, where the second hides the first

- Note that even the type can change
- Old definition is still there, but (at least at the top-level environment) it is no longer accessible

Variable identifiers

- Any “reasonable" sequence of alphanumeric characters
- We won't use things like this. . .

```
> val $$$ = "ab";
val $$$ = "ab": string
```

Examples

```
> val pi = 3.14159;
val pi = 3.14159: real
> val radius = 4.0;
val radius = 4.0: real
> pi * radius * radius;
val it = 50.26544: real
> val area = pi * radius * radius;
val area = 50.26544: real
```

$(a_1, a_2, \ldots, a_n)$

$[a_1, a_2, \ldots, a_n]$

Complex types in ML

$$(a_1, a_2, \ldots, a_n)$$

Tuples in ML

Tuples

- We have already seen something when looking at the operators
- Example

```
> val t = (1, 2, 3);
val t = (1, 2, 3): int * int * int
```

- We can mix types in tuple definitions

```
> val t = (4, 5.0, "six");
val t = (4, 5.0, "six"): int * real * string
```

- We can also use a complex type instead of a simple one

```
> val t = (1, (2, 3, 4));
val t = (1, (2, 3, 4)): int * (int * int * int)
```

Accessing tuple components:

- Let us consider

```
> val t = (4, 5.0, "six");
```

- Then

```
> #1 (t);
val it = 4: int
> #3 (t);
val it = "six": string
> #4 (t);
poly: : error: Type error in function application.
Function: #4 : 'a * 'b * 'c * 'd -> 'd
Argument: (t) : int * real * string
Reason: Can't unify {4: 'a} to int * real * string (Field 4
missing)
```

Generic type: the specific type is determined only when we use it.

$[a_1, a_2, \ldots, a_n]$

Lists in ML

Lists

- Represented with square brackets
- Example

```
> [1,2,3];
val it = [1, 2, 3]: int list
```

- Note that all elements of a list (unlike tuples) must be of the same type

More examples

```
> [1.0,2.0];
val it = [1.0, 2.0]: real list

> [1,2.0];
poly: : error: Elements in a list have different types.
Item 1: 1 : int
Item 2: 2.0 : real
Reason:
Can't unify int (*In Basis*) with real (*In Basis*)
(Different type constructors)

> [];
val it = []: 'a list
```

Generic type: the specific type is determined only when we use it.

Note that for an empty list, ML cannot determine the type of the elements

Head of a list: `hd`

- Head: first element of a list

```
> val L = [2,3,4];
val L = [2, 3, 4]: int list
> val M = [5];
val M = [5]: int list

> hd(L);
val it = 2: int
> hd(M);
val it = 5: int
```

Tail of a list: `tl`

- Tail: all the rest

```
> L;
val it = [2, 3, 4]: int list
> M;
val it = [5]: int list

> tl (L);
val it = [3, 4]: int list
> tl (M);
val it = []: int list
```

Note that ML can determine the type of this empty list

Concatenation of lists: @

- Example

```
> [1,2] @ [3,4];
val it = [1, 2, 3, 4]: int list
```

- Note that both lists must be of the same type

```
> [1,2] @ ["a","b"];
poly: : error: Type error in function application.
Function: @ : int list * int list -> int list
Argument: ([1, 2], ["a", "b"]) : int list * string list
Reason:
Can't unify int (*In Basis*) with string (*In Basis*)
(Different type constructors)
```

- Two different types of concatenation
  - ^: Strings
  - @: List

Cons: ::

- An operator that takes an element of type `'a` and a list of type `'a list` and combines them

```
> 2 :: [3,4];
val it = [2, 3, 4]: int list
> 2 :: nil;
val it = [2]: int list
> 2 :: [];
val it = [2]: int list
```

`nil` is the same as the empty list `[]`

- :: is right associative

```
> 1 :: 2 :: 3 :: nil;
val it = [1, 2, 3]: int list
> (1 :: (2:: (3::nil)));
val it = [1, 2, 3]: int list
```

- The other way would make no sense

Strings to lists: `explode`

```
> explode ("abcd");
val it = [#"a", #"b", #"c", #"d"]: char list
> explode ("");
val it = []: char list
```

Lists to strings: `implode`

```
> implode ([ #"a", #"b", #"c", #"d"]);
val it = "abcd": string
> implode (nil);
val it = "": string
> implode (explode ("xyz"));
val it = "xyz": string
```

The ML type system

- Basic types `int`, `real`, `bool`, `char`, `string`, `unit`
- Complex types: For now, 2 constructors:
  - `T1 * T2 * ... * Tn` (tuples)
  - `T list`

Examples

```
> [1, 2, 3];
val it = [1, 2, 3]: int list

> ("ab", [1,2,3], 4);
val it = ("ab", [1, 2, 3], 4): string * int
list * int

> [[(1,2),(3,4)], [(5,6)], nil];
val it = [[(1, 2), (3, 4)], [(5, 6)], []]:
(int * int) list list
```

A list of a int*int list

What’s the type … without trying it ☺

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Functions in ML

Functions

- In ML, just another type of value
- Represented by parametrized expressions
- Calculate a value based on parameters
  - No collateral effects
- Syntax `fn` (corresponds with λ in the λ-calculus, that we will see later)

```
fn <param> => <expression>;
```

- Example

```
fn n => n+1;
```

- We can directly apply the function to the parameter

```
(fn n => n+1) 5;
```

value 5 is associated to formal parameter n, and then the function is evaluated

Functions and names

- We can associate the functions to a name, just like values

```
> val increment = fn n => n+1;
val increment = fn: int -> int
```

- We also have a syntactic sugar notation for functions with names

```
> fun increment n = n+1;
val increment = fn: int -> int
```

- And then write

```
> increment 5;
val it = 6: int
```

Function types

- Example: function that converts character from lower to upper case

```
> fun upper(c) = chr (ord(c) - 32);
val upper = fn: char -> char
```

- Poly gives the type of the function. This is
  - The keyword `fn`
  - The type of the argument
  - The symbol `->`
  - The type of the result
- When using the function:

```
> upper (#"b");
val it = #"B": char
```

or

```
> upper #"b";
val it = #"B": char
```

The ML type system

- If `T1` and `T2` are types, so is

  `T1 -> T2`

- This is the type of functions that take an object of type `T1` and produce one of type `T2`
- Note that `T1` and `T2` can be any type, including function types

Specifying types

- ML deduces the types of functions automatically, as in our first example
- Another example

```
> fun square (x) = x * x;
val square = fn: int -> int
```

- But multiplication is also defined for reals. If we want to square real numbers, we can write

```
> fun square (x:real) = x * x;
val square = fn: real -> real
```

Examples

- Use of a function

```
> radius;
val it = 4.0: real
> pi;
val it = 3.14159: real
> pi * square (radius);
val it = 50.26544: real
```

- or

```
> pi * square radius;
val it = 50.26544: real
```

Multiple arguments

- All functions in ML have exactly one parameter but this parameter can be a tuple
- Example: Largest of three reals

```
> fun max3(a:real,b,c) = (* maximum of reals *)
if a>b then
if a>c then a else c
else
if b>c then b else c;
val max3 = fn: real * real * real -> real

> max3(5.0,4.0,7.0);
val it = 7.0: real
```

Note the syntax for comments

What would happen if we didn't specify that a was a real?

The input is a tuple of 3 int

```
>fun max3(a,b,c) =
  if a>b then
    if a>c then a else c
    else
      if b>c then b else c;
val max3 = fn: int * int * int -> int

> max3 (4,5,7);
val it = 7: int
```

Summary

- Type errors and conversion in ML
- Names and denotable objects and variables
- Variables in ML
- Complex types in ML

Readings

- Chapter 6 of the reference book
  - Maurizio Gabbrielli and Simone Martini "Linguaggi di Programmazione - Principi e Paradigmi", McGraw-Hill

Next time

- Type inference in ML
- Visibility rules
- Abstraction of control
- Recursion